

Email Address

testusergmail@.com

Password*

.....

Phone (optional)

! '.' is used at a wrong position in '.com'.

TCO1 A

Email Address

testuser@@mail.gcom

Password*

.....

Phone (optional)

0803385634

! A part following '@' should not contain the symbol '@'.

TCO1 B

Mayer	John
Email Adress	
testusergmail.com	
! Please include an '@' in the email address. 'testusergmail.com' is missing an '@'.	
.....	
Phone (optional)	

TC01 C

Email Adress
testuser@gmail.com
Password*

TC01 D

Schedule Meetings

You can book a meeting with us online, with just the use of your email address for us to contact you.
Get in touch with one of our staff for online on one-to-one consultation meetings.

Fields marked with * are mandatory

First Name*

Last Name*

Email

Please fill out this field.

Password*

Phone (optional)

Address*

Postcode*

Booking Type

Please select your booking

Date*

Time*

Choose Date

--:--

Book Meeting

Schedule Meetings

You can book a meeting with us online, with just the use of your email address for us to contact you.
Get in touch with one of our staff for online on one-to-one consultation meetings.

Fields marked with * are mandatory

First Name*	Last Name*
James	Roiley

Email Address

Password*

Phone (optional)

Address*	Postcode*

Bookings  Please fill out this field.

Date*	Time*
Choose Date	--:-- 

Sc Please complete bookingType gs

You can book a meeting with us online, with just the use of your email address for us to contact you.
Get in touch with one of our staff for online on one-to-one consultation meetings.

Fields marked with * are mandatory

First Name*	Last Name*
James	Roiley

Email Address

Password*

Phone (optional)

Address*	Postcode*
45 Bamber Street	JE3 EJ2

Booking Type

Date*	Time*
Choose Date	23:09

TCOB, C

Sc Please complete bookingType gs

You can book a meeting with us online, with just the use of your email address for us to contact you.
Get in touch with one of our staff for online on one-to-one consultation meetings.

Fields marked with * are mandatory

First Name*	Last Name*
James	Roiley

Email Address

Password*

Phone (optional)

Address*	Postcode*
45 Bamber Street	JE3 EJ2

Booking Type

Date*	Time*
Choose Date	23:09

TCOB, D

Sch Bookings

Booking saved successfully

You can book a meeting with us online, with just the use of your email address for us to contact you.
Get in touch with one of our staff for online on one-to-one consultation meetings.

Fields marked with * are mandatory

First Name*	Last Name*
James	Roiley
Email Address	
jamesroiley@gmail.com	
Password*	

Phone (optional)	
088088404	
Address*	Postcode*
45 Bamber Street	JE3 EJ2
Booking Type	
Installation	
Date*	Time*
April 25, 2025	23:09

LOGIN COMPONENT

Already have an account?

Log into your account to manage your bookings and schedule new appointments.

See the latest updates and news today.

Email*

Password*

!

 Please fill out this field.

Login

Don't have an account yet, make a booking request by [clicking here](#) and make your account

Already have an account?

Log into your account to manage your bookings and schedule new appointments.
See the latest updates and news today.

Email*

testuser22gmail@com,

Password*

.....

!

A part following '@' should not contain the symbol ','.

Login

Don't have an account yet, make a booking request
by [clicking here](#) and make your account

ml

RolsaTech

HomeCarbon CalculatorBook Meeting

Already have an account?

User not found

Log into your account to manage your bookings and schedule new appointments.
See the latest updates and news today.

Email*

testuser@gmail.com

Password*

Login

Don't have an account yet, make a booking request
by [clicking here](#) and make your account

TC02, C

USER AUTHENTICATION

Already Invalid Credentials account?

Log into your account to manage your bookings and schedule new appointments.
See the latest updates and news today.

Email*

rakeljohn@gmail.com

Password*

.....

Login

Don't have an account yet, make a booking request
by [clicking here](#) and make your account

TC03 A (Invalid password as password123)

0/pages/login.html

RolsaTech Home Carbon Calculator Book Meeting

Already Authentication Successful count?

Log into your account to manage your bookings and schedule new appointments.
See the latest updates and news today.

Email*
rakiljohn@gmail.com

Password*

Login

Don't have an account yet, make a booking request
by [clicking here](#) and make your account

TC03, B(Valid password as rakeljohn)

olsa Tech x +

localhost:8080/dashboard?id=12

RolsaTech Latest Updates Log Out New bookings

Welcome back, Rakel

Current Bookings

Location: 80 Road Block

Postcode: FY2 3FE

Booking Type: Installation

Scheduled at: Friday, April 25, 2025 at 12:00:00 AM

TC03, C(Use successfully redirect to dashboard, see server logs below)

```

DST /login 401 1.171 ms - 28
OPTIONS /login 204 0.141 ms - 0
login post endpoint]
undefined
DST /login 401 1.171 ms - 28
OPTIONS /login 204 0.118 ms - 0
login post endpoint]
undefined
DST /login 401 1.306 ms - 28
OPTIONS /login 204 0.152 ms - 0
login post endpoint]

id: 12,
password: '$2b$10$7GCCP42Xc30ffQLTyCpAuuFM/KSZnLun2s0b.DWaCP8m9F3BaVSfC'

passwords match
DST /login 200 60.090 ms - 60
dashboard endpoint]
ET /dashboard?id=12 304 1.702 ms - -
ET /dashboard.js 304 0.555 ms - -
time: '12:30:00',
firstLogin: 1,
createdAt: 2025-04-23T08:59:57.000Z

DST /api/data 200 1.686 ms - 301
OPTIONS /login 204 0.118 ms - 0
login post endpoint]

id: 12,
password: '$2b$10$7GCCP42Xc30ffQLTyCpAuuFM/KSZnLun2s0b.DWaCP8m9F3BaVSfC'

DST /login 401 58.317 ms - 33
OPTIONS /login 204 0.127 ms - 0
login post endpoint]

id: 12,
password: '$2b$10$7GCCP42Xc30ffQLTyCpAuuFM/KSZnLun2s0b.DWaCP8m9F3BaVSfC'

passwords match
DST /login 200 64.091 ms - 60
dashboard endpoint]
ET /dashboard?id=12 304 4.960 ms - -
ET /dashboard.js 304 0.620 ms - -
ET /base.css 304 2.652 ms - -
ET /dashboard.css 304 3.039 ms - -
id: '12' }
2

id: 12,
firstName: 'Rake1',
lastName: 'John',
email: 'rake1john@gmail.com',
type: 'Installation',
phone: null,

```

API TESTING WITH JEST

TCO4's

```
5 endpoints.test.js | describe('Testing authentication system with valid credentials') callback { // test 'Expect status code returned to be 200' callback
  6   const axiosResponse = await axios.get('http://localhost:5500/api/auth/login')
  7 }
  8
  9 describe('Testing dashboard page', () => {
 10   test('Expect status code returned to be 200', async () => {
 11     const response = await axios.get(dashboardEndpoint)
 12
 13     expect(response.status).toBe(200)
 14   })
 15 })
 16
 17 describe('Testing home on port 5500 page', () => {
 18   test('Expect status code returned to be 200', async () => {
 19     const response = await axios.get(indexPage)
 20
 21     expect(response.status).toBe(200)
 22     console.log(response.headers)
 23   })
 24 })
 25
 26 describe('Testing login on port 5500 ', () => {
 27   test('Expect status code returned to be 200', async () => {
 28     const response = await axios.get(loginPage)
 29
 30     expect(response.status).toBe(200)
 31   })
 32 })
 33
 34 describe('Testing booking page on port 5500 ', () => {
 35   test('Expect status code returned to be 200', async () => {
 36     const response = await axios.get(bookingPage)
 37
 38     expect(response.status).toBe(200)
 39   })
 40 })
 41
 42 describe('Testing quiz page on port 5500 ', () => {
 43   test('Expect status code returned to be 200', async () => {
 44
 45   })
 46 })
 47
 48 describe('Testing quiz page on port 5500 ', () => {
 49   test('Expect status code returned to be 200', async () => {
 50
 51   })
 52 })
 53
 54 describe('Testing quiz page on port 5500 ', () => {
 55   test('Expect status code returned to be 200', async () => {
 56
 57   })
 58 })
 59
 60 }
 61
 62 at Object.log (endpoints.test.js:131:21)
 63
 64 PASS ./endpoints.test.js
 65   Testing dashboard page
 66     ✓ Expect status code returned to be 200 (37 ms)
 67   Testing home on port 5500 page
 68     ✓ Expect status code returned to be 200 (20 ms)
 69   Testing login on port 5500
 70     ✓ Expect status code returned to be 200 (3 ms)
 71   Testing booking page on port 5500
 72     ✓ Expect status code returned to be 200 (2 ms)
 73   Testing quiz page on port 5500
 74     ✓ Expect status code returned to be 200 (2 ms)
 75   Testing authentication system with empty data
 76     ✓ Expect server response to be 401 (56 ms)
 77   Testing authentication system with a non-existing user
 78     ✓ Expect server response to be 401 (6 ms)
 79   Testing authentication system with invalid credentials
 80     ✓ Expect server response to be 401 (57 ms)
 81   Testing authentication system with valid credentials
 82     ✓ Expect server response to be 200 (56 ms)
 83
 84 Test Suites: 1 passed, 1 total
 85 Tests: 9 passed, 9 total
 86 Snapshots: 0 total
 87 Time: 0.619 s, estimated 1 s
 88 Ran all test suites.
 89
 90 user@DESKTOP-V7HENL8 MINGW64 ~/Desktop/tests (master)
 91
 92 describe('Testing authentication system with valid credentials') callback { // test 'Expect status code returned to be 200' callback
 93   const axiosResponse = await axios.get('http://localhost:5500/api/auth/login')
 94 }
 95
 96 describe('Testing dashboard page', () => {
 97   test('Expect status code returned to be 200', async () => {
 98     const response = await axios.get(dashboardEndpoint)
 99
100     expect(response.status).toBe(200)
101   })
102 })
103
104 describe('Testing home on port 5500 page', () => {
105   test('Expect status code returned to be 200', async () => {
106     const response = await axios.get(indexPage)
107
108     expect(response.status).toBe(200)
109     console.log(response.headers)
110   })
111 })
112
113 describe('Testing login on port 5500 ', () => {
114   test('Expect status code returned to be 200', async () => {
115     const response = await axios.get(loginPage)
116
117     expect(response.status).toBe(200)
118   })
119 })
120
121 describe('Testing booking page on port 5500 ', () => {
122   test('Expect status code returned to be 200', async () => {
123     const response = await axios.get(bookingPage)
124
125     expect(response.status).toBe(200)
126   })
127 })
128
129 describe('Testing quiz page on port 5500 ', () => {
130   test('Expect status code returned to be 200', async () => {
131
132   })
133 })
134
135 describe('Testing quiz page on port 5500 ', () => {
136   test('Expect status code returned to be 200', async () => {
137
138   })
139 })
140
141 describe('Testing quiz page on port 5500 ', () => {
142   test('Expect status code returned to be 200', async () => {
143
144   })
145 })
146
147 }
148
149 at Object.log (endpoints.test.js:131:21)
150
151 PASS ./endpoints.test.js
152   Testing dashboard page
153     ✓ Expect status code returned to be 200 (34 ms)
154   Testing home on port 5500 page
155     ✓ Expect status code returned to be 200 (20 ms)
156   Testing login on port 5500
157     ✓ Expect status code returned to be 200 (3 ms)
158   Testing booking page on port 5500
159     ✓ Expect status code returned to be 200 (2 ms)
160   Testing quiz page on port 5500
161     ✓ Expect status code returned to be 200 (2 ms)
162   Testing authentication system with empty data
163     ✓ Expect server response to be 401 (56 ms)
164   Testing authentication system with a non-existing user
165     ✓ Expect server response to be 401 (5 ms)
166   Testing authentication system with invalid credentials
167     ✓ Expect server response to be 401 (57 ms)
168   Testing authentication system with valid credentials
169     ✓ Expect server response to be 200 (56 ms)
170
171 Test Suites: 1 passed, 1 total
172 Tests: 9 passed, 9 total
173 Snapshots: 0 total
174 Time: 0.619 s, estimated 1 s
175 Ran all test suites.
176
177 user@DESKTOP-V7HENL8 MINGW64 ~/Desktop/tests (master)
178
179 describe('Testing authentication system with a non-existing user', () => {
180   test('Expect status code returned to be 200', async () => {
181     const response = await axios.get('http://localhost:5500/api/auth/login')
182
183     expect(response.status).toBe(200)
184   })
185 })
186
187 // testing on existing user with invalid credentials
188 describe('Testing authentication system with invalid credentials', () => {
189   test('Expect server response to be 401', async () => {
190     const response = await axios.get('http://localhost:5500/api/auth/login')
191
192     expect(response.status).toBe(401)
193   })
194 })
195
196 at Object.log (endpoints.test.js:26:17)
197
198 console.log
199   Object {
200     'x-powered-by': 'Express',
201     'access-control-allow-origin': 'http://localhost:5500',
202     vary: 'Origin',
203     'content-type': 'application/json; charset=utf-8',
204     'content-length': '60',
205     etag: 'W/"3c-MDRQonQvAE3BwSunb132hNS21g"',
206     date: 'Wed, 23 Apr 2025 14:10:58 GMT',
207     connection: 'keep-alive',
208     'keep-alive': 'timeout=5'
209   }
210
211 describe
212   at Object.log (endpoints.test.js:131:21)
213
214 test
215   at Object.log (endpoints.test.js:131:21)
216
217 PASS ./endpoints.test.js
218   Testing dashboard page
219     ✓ Expect status code returned to be 200 (34 ms)
220   Testing home on port 5500 page
221     ✓ Expect status code returned to be 200 (20 ms)
222   Testing login on port 5500
223     ✓ Expect status code returned to be 200 (3 ms)
224   Testing booking page on port 5500
225     ✓ Expect status code returned to be 200 (2 ms)
226   Testing quiz page on port 5500
227     ✓ Expect status code returned to be 200 (2 ms)
228   Testing authentication system with empty data
229     ✓ Expect server response to be 401 (56 ms)
230   Testing authentication system with a non-existing user
231     ✓ Expect server response to be 401 (5 ms)
232   Testing authentication system with invalid credentials
233     ✓ Expect server response to be 401 (57 ms)
234   Testing authentication system with valid credentials
235     ✓ Expect server response to be 200 (56 ms)
236
237 Test Suites: 1 passed, 1 total
238 Tests: 9 passed, 9 total
239 Snapshots: 0 total
240 Time: 0.619 s, estimated 1 s
241 Ran all test suites.
242
243 user@DESKTOP-V7HENL8 MINGW64 ~/Desktop/tests (master)
244
245 describe('Testing authentication system with a non-existing user', () => {
246   test('Expect status code returned to be 200', async () => {
247     const response = await axios.get('http://localhost:5500/api/auth/login')
248
249     expect(response.status).toBe(200)
250   })
251 })
252
253 // testing on existing user with invalid credentials
254 describe('Testing authentication system with invalid credentials', () => {
255   test('Expect server response to be 401', async () => {
256     const response = await axios.get('http://localhost:5500/api/auth/login')
257
258     expect(response.status).toBe(401)
259   })
260 })
261
262 at Object.log (endpoints.test.js:26:17)
263
264 console.log
265   Object {
266     'x-powered-by': 'Express',
267     'access-control-allow-origin': 'http://localhost:5500',
268     vary: 'Origin',
269     'content-type': 'application/json; charset=utf-8',
270     'content-length': '60',
271     etag: 'W/"3c-MDRQonQvAE3BwSunb132hNS21g"',
272     date: 'Wed, 23 Apr 2025 14:10:58 GMT',
273     connection: 'keep-alive',
274     'keep-alive': 'timeout=5'
275   }
276
277 describe
278   at Object.log (endpoints.test.js:131:21)
279
280 test
281   at Object.log (endpoints.test.js:131:21)
282
283 PASS ./endpoints.test.js
284   Testing dashboard page
285     ✓ Expect status code returned to be 200 (34 ms)
286   Testing home on port 5500 page
287     ✓ Expect status code returned to be 200 (20 ms)
288   Testing login on port 5500
289     ✓ Expect status code returned to be 200 (3 ms)
290   Testing booking page on port 5500
291     ✓ Expect status code returned to be 200 (2 ms)
292   Testing quiz page on port 5500
293     ✓ Expect status code returned to be 200 (2 ms)
294   Testing authentication system with empty data
295     ✓ Expect server response to be 401 (56 ms)
296   Testing authentication system with a non-existing user
297     ✓ Expect server response to be 401 (5 ms)
298   Testing authentication system with invalid credentials
299     ✓ Expect server response to be 401 (57 ms)
300   Testing authentication system with valid credentials
301     ✓ Expect server response to be 200 (56 ms)
302
303 Test Suites: 1 passed, 1 total
304 Tests: 9 passed, 9 total
305 Snapshots: 0 total
306 Time: 0.619 s, estimated 1 s
307 Ran all test suites.
308
309 user@DESKTOP-V7HENL8 MINGW64 ~/Desktop/tests (master)
310
311 describe('Testing authentication system with a non-existing user', () => {
312   test('Expect status code returned to be 200', async () => {
313     const response = await axios.get('http://localhost:5500/api/auth/login')
314
315     expect(response.status).toBe(200)
316   })
317 })
318
319 // testing on existing user with invalid credentials
320 describe('Testing authentication system with invalid credentials', () => {

```

```

80     password: "password1234"
81   }
82
83   describe("Testing authentication system with a non-existing user", () => {
84     test("Expect server response to be 401", async () => {
85       try {
86         const response = await axios.post(loginEndpoint, {
87           // testUser
88           email: "testuser001@gmail.com",
89           password: "password1234",
90         })
91       } catch(err){
92         expect(err.response.status).toBe(401)
93         expect(err.response.data.message).toBe("User not found")
94       }
95     })
96   })
97 })
98
99
100 // testing an existing user with invalid credentials
101 describe("Testing authentication system with invalid credentials", () => {

```

```

  ✓ Expect status code returned to be 200 (2 ms)
  Testing authentication system with empty data
  ✓ Expect server response to be 401 (56 ms)
  Testing authentication system with a non-existing user
  ✓ Expect server response to be 401 (6 ms)
  Testing authentication system with invalid credentials
  ✓ Expect server response to be 401 (57 ms)
  Testing authentication system with valid credentials

```

```

96     })
97   })
98
99
100   // testing an existing user with invalid credentials
101   describe("Testing authentication system with invalid credentials", () => {
102     test("Expect server response to be 401", async () => {
103       try {
104         const response = await axios.post(loginEndpoint, {
105           email: "rakeljohn@gmail.com",
106           password: "password1234",
107         })
108       } catch (err) {
109         expect(err.response.status).toBe(401)
110         expect(err.response.data.message).toBe("Invalid Credentials")
111       }
112     })
113   })
114 })
115
116 const validUser = {
117   email: "rakeljohn@gmail.com",
118   password: "rakeljohn"
119 }
120 // testing with valid credentials, a redirect url should be sent in HTTP header
121 describe("Testing authentication system with valid credentials", () => {
122   test("Expect server response to be 200", async () => {
123     const response = await axios.post(loginEndpoint, {
124       email: "rakeljohn@gmail.com",
125       password: "rakeljohn",
126     })
127
128     expect(response.status).toBe(200)
129     // this means that the user was successfully redirected to the dashboard page
130     // expect(response.headers["content-type"]).toMatch(/text\/html/)
131     console.log(response.headers)

```

```
123     const response = await axios.post(loginEndpoint, {
124       email: "rakiljohn@gmail.com",
125       password: "rakiljohn",
126     })
127
128     expect(response.status).toBe(200)
129     // this means that the user was succesfully redirected to the dashboard page
130     // expect(response.headers["content-type"]).toMatch(/text\/html/)
131     console.log(response.headers)
132
133   })
134 })
```

TERMINAL PROBLEMS OUTPUT DEBUG CONSOLE PORTS

PASS ./endpoints.test.js

- Testing dashboard page
 - ✓ Expect status code returned to be 200 (37 ms)
- Testing home on port 5500 page
 - ✓ Expect status code returned to be 200 (20 ms)
- Testing login on port 5500
 - ✓ Expect status code returned to be 200 (3 ms)
- Testing booking page on port 5500
 - ✓ Expect status code returned to be 200 (4 ms)
- Testing quiz page on port 5500
 - ✓ Expect status code returned to be 200 (2 ms)
- Testing authentication system with empty data
 - ✓ Expect server response to be 401 (56 ms)
- Testing authentication system with a non-existing user
 - ✓ Expect server response to be 401 (6 ms)
- Testing authentication system with invalid credentials
 - ✓ Expect server response to be 401 (57 ms)
- Testing authentication system with valid credentials
 - ✓ Expect server response to be 200 (61 ms)

Test Suites: 1 passed, 1 total
Tests: 9 passed, 9 total

UI TESTING

address for us to contact you. Get in touch with one of our staff for
online on one-to-one consultation meetings.

Fields marked with * are mandatory

label 358.09 × 25
Color ■ #252424
Font 16px Poppins, Arial, Helvetica, sans-serif
Margin 3.57812px 0px 0px

ACCESSIBILITY
Contrast Aa 12.54 ✓
Name
Role
Keyboard-focusable

Password*

Phone (optional)

088088404

Address* Postcode*

45 Bamber Street IE3 5 12

Phone (optional)

088088404

Address* Postcode*

45 Bamber Street

Booking Type

Installation

Date*

Choose

button 194.56 × 47.78
Color ■ #102910
Font 21.76px Poppins, Arial, Helvetica, sans-serif
Background □ #FFFFFF
Margin 7.15625px 0px 0px
Padding 6.4px 23.36px

ACCESSIBILITY
Contrast Aa 15.59 ✓
Name Book Meeting
Role button
Keyboard-focusable ✓

Book Meeting

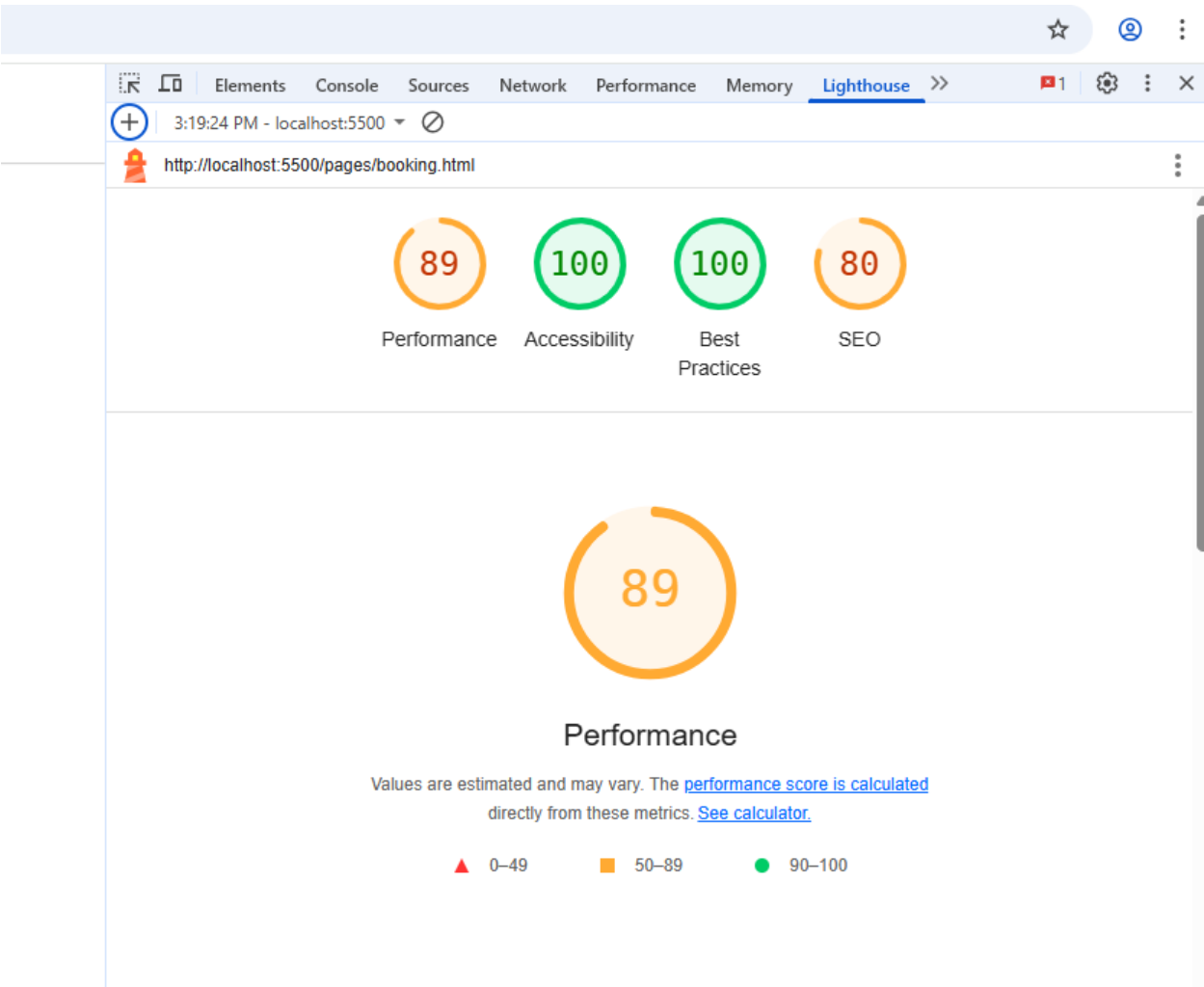
Schedule Meetings

You can book a meeting with us online, with just the use of your email address for us to contact you. Get in touch with one of our staff for online on one-to-one consultation meetings.

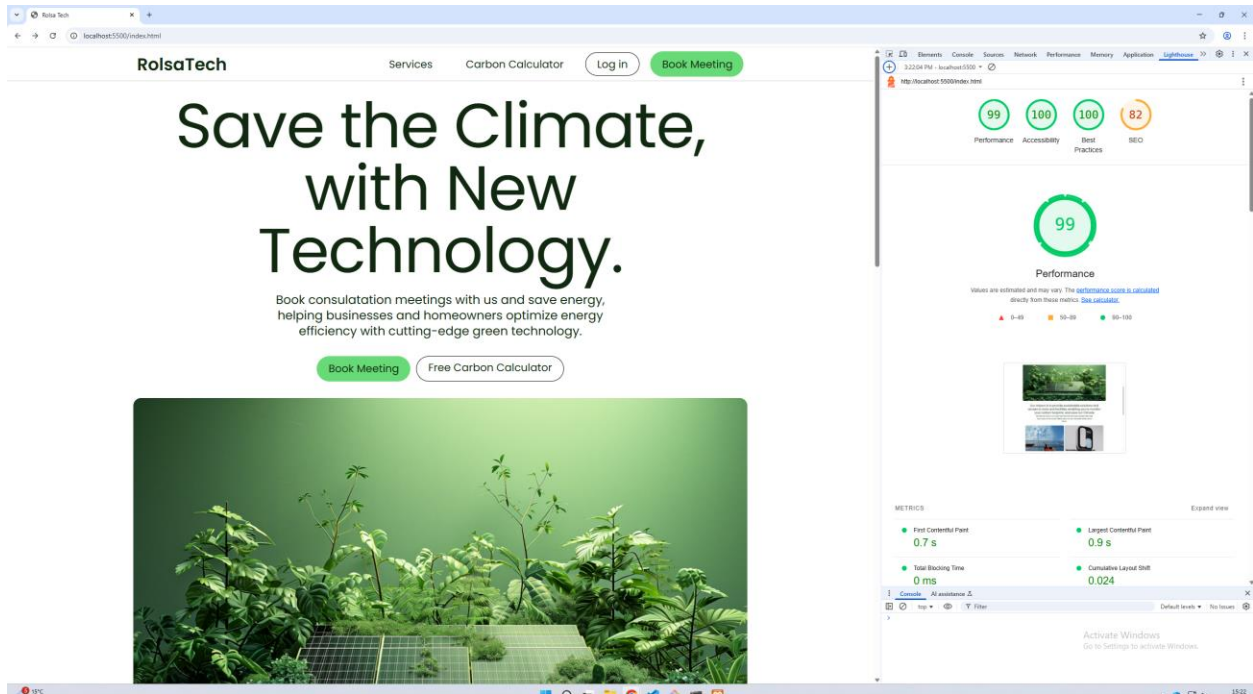
The screenshot shows a web form titled "Schedule Meetings" with the following fields: "Email Address" (containing "jamesriley@gmail.com"), "Password*" (masked with dots), "Phone (optional)" (containing "088088404"), "Address*" (containing "45 Bamber Street"), and "Postcode*" (containing "JE3 3J2"). An accessibility overlay is visible in the top left corner, displaying the following information:

- label: 358.09 x 25
- Color: #252424
- Font: 16px Poppins, Arial, Helvetica, sans-serif
- Margin: 3.57812px 0px 0px
- ACCESSIBILITY: Contrast: 12.54 (checked), Name, Role, Keyboard-focusable

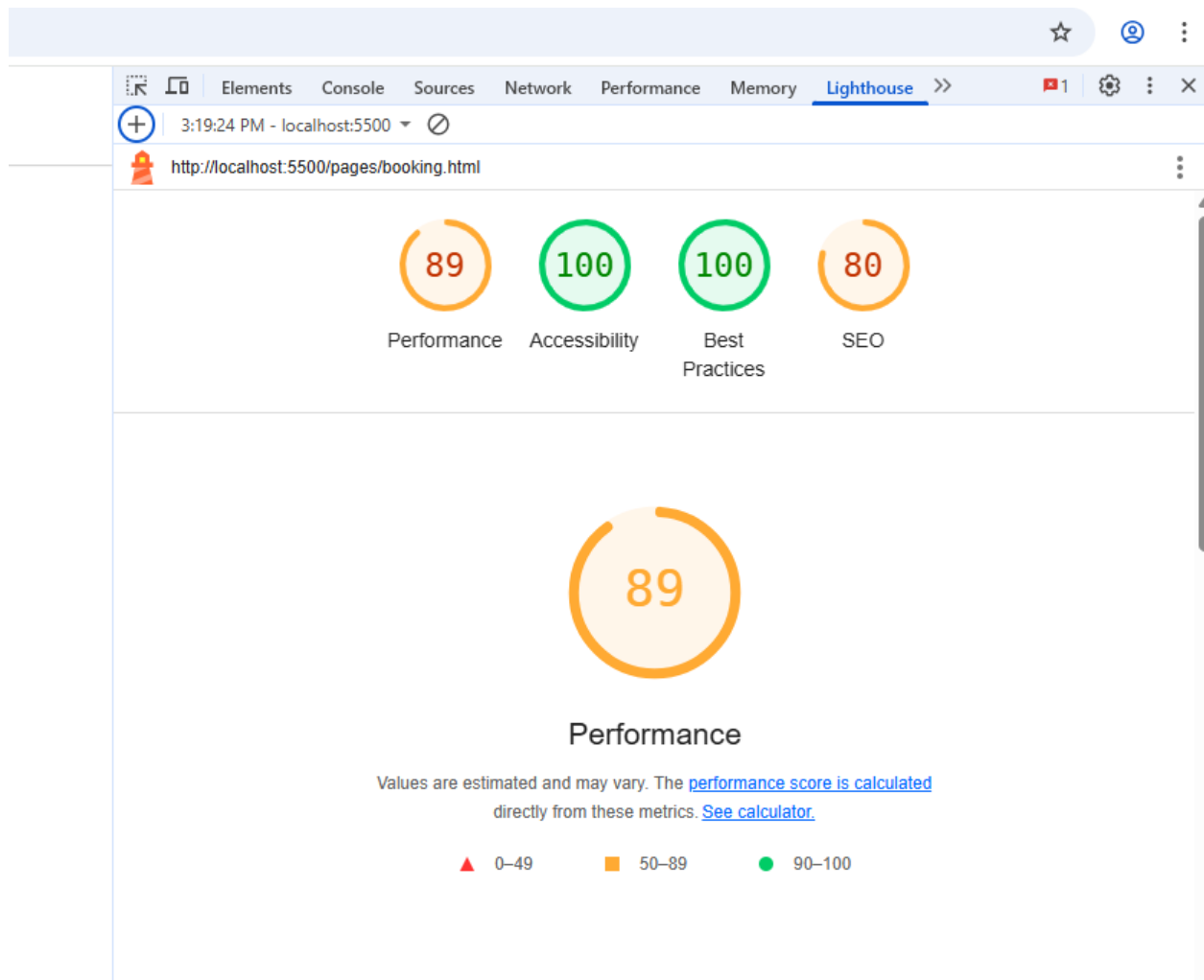
LIGHTHOUSE AUDITS



TC06



PERFORMANCE METRICS AND SEO METRICS



TC07

SERVER RESPONSE TIME TC09

```
[nodemon] watching extensions: .js, .mjs, .json
[nodemon] starting 'node server.js'
[listening...] on port 8080
[dashboard endpoint]
GET /dashboard 200 9.178 ms - 4155
[login post endpoint]
POST /login 401 0.981 ms - 25
[login post endpoint]
undefined
POST /login 401 20.058 ms - 28
[login post endpoint]
{
  id: 12,
```